

MATH 220, Matrices
Penn State University, University Park
Fall 2026

COURSE INFORMATION

CATALOG DESCRIPTION: MATH 220 (GQ) MATRICES (3 credits) Systems of linear equations appear everywhere in mathematics and its applications. MATH 220 will give students the basic tools necessary to analyze and understand such systems. The initial portion of the course teaches the fundamentals of solving linear systems. This requires the language and notation of matrices and fundamental techniques for working with matrices such as row and column operations, echelon form, and invertibility. The determinant of a matrix is also introduced; it gives a test for invertibility. In the second part of the course the key ideas of eigenvector and eigenvalue are developed. These allow one to analyze a complicated matrix problem into simpler components and appear in many disguises in physical problems. The course also introduces the concept of a vector space, a crucial element in future linear algebra courses.

PREREQUISITE: MATH 110 or MATH 140 or MATH 140B or MATH 140E or MATH 140G or MATH 140H

TEXTBOOK: MyLab Math with Pearson eText -- Instant Access -- for Linear Algebra and its Applications (18-Weeks) ISBN-13: 9780135851203. It is easiest if you purchase it through our Canvas course.

COURSE FORMAT: There are three 50-minute lectures each week. See the calendar at the end of this syllabus.

IMPORTANT COURSE DATES:

- Midterm Examinations: Tuesday, September 29 and Wednesday, April 1
 - Makeup Midterm Examinations: Thursday, November 5 and Monday, November 9
 - Final Examination: During the week of finals (December 14-18)
- See the Assessments section below for more information on examinations.

COURSE GOALS AND OBJECTIVES

COURSE GOALS

1. Analyze and solve a system of linear equations.
2. Understand the important characteristics of matrices such as its fundamental subspaces, rank, determinant, eigenvalues, and eigenvectors.
3. Learn how to use characteristics of a matrix to solve a system of linear equations or study properties of a linear transformation.
4. Understand important concepts of vector spaces such as independence, basis, dimension and orthogonality.
5. Use linear algebra terminology to express problems in branches of science; communicate solutions back in terms of the original context.
6. Apply linear algebra to computationally robust problems using Mathematica or another computing program.

COURSE OBJECTIVES

Section 1.1 – Systems of Linear Equations

1. Use elementary row operations to solve systems of linear equations.
2. Determine if a system of linear equations is consistent.
3. Determine the conditions for which a linear system is consistent.
4. Determine the validity of statements about systems of linear equations, row operations, or matrices.

Section 1.2 – Row Reduction and Echelon Forms

1. Identify matrices in echelon form and reduced echelon form.
2. Row reduce matrices to reduced echelon form.
3. Find the general solution to a system with a given augmented matrix.
4. Determine if a solution is consistent given a description of the corresponding coefficient matrix.

5. Determine the conditions for which a linear system has specified types of solutions.
6. Determine the validity of statements about row reduction and echelon forms.
7. Find interpolating polynomials for small data sets using row reduction.
8. Solve applications using row reduction.

Section 1.3 – Vector Equations

1. Compute sums and scalar products of vectors, both algebraically and geometrically.
2. Convert between vector equations and systems of equations.
3. Determine if a vector is a linear combination of other vectors.
4. Characterize the span of a set of vectors algebraically or geometrically.
5. Determine the validity of statements about vectors and vector equations.
6. Solve applications involving vector equations.

Section 1.4 – The Matrix Equation $Ax=b$

1. Compute the product of a matrix and a vector.
2. Convert between matrix equations, vector equations, and systems of equations.
3. Solve matrix equations using augmented matrices.
4. Characterize the span of the column vectors of a matrix.
5. Determine whether a matrix equation has no solution, one solution, or infinitely many solutions.
6. Determine the validity of statements about vector equations and matrix equations.
7. Solve applications involving matrix equations.

Section 1.5 – Solution Sets of Linear Systems

1. Determine if a system of equations has a nontrivial solution.
2. Solve a system of equations or a matrix equation and write the solution in parametric form.
3. Describe the solution sets of systems of equations geometrically.
4. Determine the validity of statements about solution sets of linear equations.

Section 1.6 – Applications of Linear Systems

1. Setup and solve linear difference equations representing changes in population.
2. Describe and solve network flow diagrams using linear systems.
3. Use vector equations to balance chemical equations.
4. Interpret solutions to applications involving linear systems.

Section 1.7 – Linear Independence

1. Determine if a set of vectors is linearly independent and determine if a vector is in a given span.
2. Determine conditions for which vectors are linearly independent or have a given span.
3. Determine the validity of statements about linear independence.

Section 1.8 – Introduction to Linear Transformations

1. Algebraically find the image of a given vector under a linear transformation.
2. Given a linear transformation $T(x)=Ax$, find x for a given b in the image of T .
3. Determine the conditions for which a linear transformation has a given domain and codomain.
4. Determine if a vector is in the range of a linear transformation.
5. Geometrically describe the image of a vector under a linear transformation.
6. Use the linearity of transformations to find the images of vectors under the transformation.
7. Determine the validity of statements about linear transformations.

Section 1.9 – The Matrix of a Linear Transformation

1. Find the standard matrix of a linear transformation.
2. Find vectors whose images under a linear transformation are given.
3. Determine the validity of statements about properties of linear transformations.
4. Determine if linear transformations are one-to-one or onto.

Section 2.1 – Matrix Operations

1. Compute sums, products, and scalar products of matrices.
2. Find values of matrices such that products of matrices have given properties.
3. Determine the validity of statements about matrix operations.

Section 2.2 – The Inverse of a Matrix

1. Find the inverse of a 2×2 matrix using the formula.
2. Use the inverse of a matrix to solve a linear system.
3. Determine the validity of statements about inverses of matrices.
4. Solve equations involving invertible matrices.
5. Find the inverse of a matrix using row reduction.

Section 2.3 – Characterizations of Invertible Matrices

1. Use the Invertible Matrix Theorem to determine the validity of statements about matrices.
2. Find the inverse of a linear transformation.

Section 3.1 – Introduction to Determinants

1. Compute determinants using cofactor expansions.
2. Determine the effect of elementary row operations on a determinant.
3. Calculate determinants of scalar multiples of matrices.

Section 3.2 – Properties of Determinants

1. Identify properties of determinants
2. Find determinants by row reduction to echelon forms.
3. Use properties of determinants to evaluate determinants.
4. Prove properties of determinants.
5. Use determinants to determine if a matrix is invertible or a set of vectors is linearly independent.
6. Determine the validity of statements about properties of determinants.

Section 4.1 – Vector Spaces and Subspaces

1. Determine whether a given set is a vector space.
2. Determine whether a given set is a subspace.
3. Determine the validity of statements about vector spaces and subspaces.
4. Solve applications involving vector spaces and subspaces.

Section 4.2 – Null Spaces, Column Spaces, and Linear Transformations

1. Determine whether a vector is in the null or column space of a matrix.
2. Find a nonzero vector in the null, column, or row space of a matrix.
3. Determine the validity of statements about linear transformations and null, column, and row spaces.
4. Use the theory of null and column spaces to find solutions of linear systems.

Section 4.3 – Linearly Independent Sets, Bases

1. Determine whether a set is linearly independent, or is a basis for, or spans a vector space.
2. Find bases for the null, column, or row space of a matrix.
3. Find a basis for the span of a set of vectors.
4. Determine the validity of statements about linearly independent sets and bases.

Section 4.5 – The Dimension of a Vector Space

1. Find the dimension of a subspace.
2. Find the dimensions of the null, column, and row spaces for a matrix.

Section 5.1 – Eigenvectors and Eigenvalues

1. Determine if a vector or number is an eigenvector or eigenvalue of a given matrix.
2. Find a basis for the eigenspace corresponding to an eigenvalue.
3. Find the eigenvalues of matrices.
4. Determine the validity of statements about eigenvectors and eigenvalues.

Section 5.2 – The Characteristic Equation

1. Find the characteristic polynomial and eigenvalues of a 2×2 matrix.
2. Find the characteristic polynomial of a 3×3 matrix.
3. Find the eigenvalues of triangular matrices.

Section 5.3 – Diagonalization

1. Compute A^k for $A = PDP^{-1}$.
2. Use the diagonalization theorem to find the eigenvalues of a matrix and a basis for each eigenspace.

3. Diagonalize matrices.
4. Determine the validity of statements about diagonalization.
5. Determine whether matrices are diagonalizable.
6. Answer conceptual questions about diagonalization of matrices.

Section 6.1 – Inner Product, Length and Orthogonality

1. Perform operations on vectors using inner products.
2. Find a unit vector.
3. Find the distance between two vectors.
4. Determine whether two vectors are orthogonal.

Section 6.2 – Orthogonal Sets

1. Determine whether a set of vectors is orthogonal.
2. Find an orthogonal projection onto a line through a given vector and the origin.
3. Write a vector as a sum of two orthogonal vectors.
4. Find the distance between a vector and a line through the origin.
5. Determine whether a set of vectors is orthonormal.
6. Determine the validity of statements about orthogonal sets or matrices and orthogonal projections.

Section 6.3 – Orthogonal projections

1. Find the orthogonal decomposition of a vector.
2. Find an orthogonal projection onto a given subspace.
3. Use the best approximation theorem to find the closest point or a distance.
4. Construct a vector that is orthogonal to a given orthogonal set.
5. Determine the validity of statements about orthogonal projections onto subspaces.

Section 6.4 – The Gram-Schmidt Process

1. Use the Gram-Schmidt process to produce an orthogonal basis and find an orthonormal basis.

Section 6.5 – Least-Squares Problems

1. Use normal equations to find least-squares solutions of systems.
2. Determine the validity of statements about least-squares solutions.

Section 7.1 – Diagonalization of Symmetric Matrices

1. Determine the validity of statements about symmetric matrices.
2. Use the Spectral Theorem for Symmetric Matrices to find a spectral decomposition of symmetric matrices.

Section 7.4 – The Singular Value Decomposition

1. Find the singular values of a matrix.
2. Find a singular value decomposition of a matrix.
3. Use the singular value decomposition of a matrix to find particular values or vectors.

This course satisfies the General Education learning objectives Key Literacies and Critical & Analytical Thinking.

GETTING HELP OUTSIDE OF CLASS

OFFICE HOURS: Weekly office hours are available. See Canvas for a schedule of hours.

TUTORS and PENN STATE LEARNING: Free mathematics tutoring is available at Penn State Learning. You can find information [here](#) about their online and in-person options.

LA SESSIONS: Each week learning assistants (LAs), who are part of the instructional team, will hold multiple evening problem sessions. These sessions are a continuation of what is discussed in class each week. By attending these sessions, you are strengthening your skills and understanding the content. More information about the general structure, time, and location of sessions will be communicated via Canvas.

ASSESSMENTS

HOMEWORK ASSIGNMENTS: There are two types of homework assignments. Written homework problem sets will be assigned by your instructor. Credit is given for writing full and complete solutions to the problems assigned. A scanned copy of the completed work must be submitted through your Gradescope account with pages assigned.

Online homework assignments are completed through your MyLab Math account. You are responsible for purchasing and activating your MyLab Math account no later than Friday, August 28th.

LATE HOMEWORK ASSIGNMENT POLICY: Due dates for every assignment are provided on Canvas. Unless otherwise stated, assignments are due on those dates. Students must notify their instructor via email with a valid excuse within 24 hours of an assignment due date to request an extension. The instructor will review late requests and circumstances on a case-by-case basis and make decisions accordingly. Late assignments will only be accepted with prior permission.

QUESTIONS REGARDING ASSIGNMENT GRADES: Questions regarding the grading of homework assignments should be directed to your section instructor within two weeks of the date the assignment is graded. Questions regarding the grading of exams or missing exam grades should be directed to your section instructor as soon as possible. Students and instructors will have **until the next exam** to review a physical copy of the exam or report a missing grade. Students will have one week from the date of the final exam to contact their instructor regarding questions about the final exam or a missing final exam grade.

MIDTERM EXAMINATIONS: Two 75-minute evening examinations will be given during the semester and a 110-minute comprehensive final exam will be given during the final exam period. No books, notes, or calculators may be used on the examinations. You must bring your University ID card to all exams. The midterm examinations will be held from **6:15 to 7:30 p.m.** on the following dates:

Midterm 1	Tuesday, September 29
Midterm 2	Thursday, November 5

Rooms for examinations will be announced by your instructor later. It is not permissible to take the exam in a different instructor's assigned room. A deduction may be assessed for not taking the test where assigned.

FINAL EXAMINATION: The final examination will be given during finals week, December 14-18. The final examination may be scheduled on any day during the final examination period. **Do not plan to leave University Park until after Friday, December 18.** Students may access their final exam schedules on Monday, September 28th through their LionPATH account. Notification of conflicts is given on the student's final exam schedule. Final exam conflicts must be submitted through Lionpath Monday, September 28th through Sunday, October 25th.

EXAM INFORMATION

Student Instructions for Exams

- We will not be using the check-in system before each exam.
- Prior to the start of any exam, students must remove all headphones, earbuds, any type of watch (including non-smartwatches), and any other electronic devices that they have on/with them.
- Students are NOT permitted to use a calculator or any electronic assistance.
- All items must be placed in a backpack or bag. Students without a backpack or bag must surrender these items to the exam proctors, who will keep the items at the front of the room. **Phones may NOT remain in pockets.**
- Students should place backpacks and bags in the row in front of their seat. To allow space for bags, students should avoid sitting in the front row of the room or seat section.
- All students must bring a **physical photo ID** (Penn State ID, driver's license, passport, or other government-issued photo identification) for identity verification. You may NOT use your phone to access your ID.
- Students may not have scratch paper, cheat sheets, note cards, or formula sheets for their exam. No documents or written information from the exam may leave the exam room with the student.

Who may take the Makeup Exam without a penalty? Students may take the make-up exam who have a university-approved documented reason such as: official class conflict, evening exams for other courses, religious observances that occur on the date *and* time of the scheduled exam and required travel for a university-sponsored activity. Students must be prepared to provide documentation. However, providing documentation does not guarantee approval. ***Students may be granted at most one makeup exam without penalty during the semester for a non-university-approved absence.***

Who may take the Makeup Exam with a penalty? Students who miss the exam may take the makeup exam for Exam 1 or Exam 2 with a 20-point penalty for reasons such as: personal business (travel, employment, weddings, graduations,...), attendance at public events (concerts, sporting events, and Greek Rush events,...), or forgetting exam details (date, time, or room)

Who may NOT take the Makeup Exam? Students who have already took the Regular exam are NOT permitted to take the Makeup exam. Doing so will result in a score of zero and may constitute an Academic Integrity violation.

Makeup Exam Policies Students who miss both the Regular exam and the Makeup exam without an approved excuse will receive a score of zero.

To be eligible for a Makeup exam students must:

1. E-mail their instructor to say that they're signing up for the Makeup exam and provide the reason for the request.
2. Submit a makeup exam request using the department's online form by the published deadline in this syllabus.
3. Provide any documentation supporting their request.

How and when to request a Makeup exam

1. Students must submit a makeup exam request through the Math Department's online request form. The student must include the reason for needing a makeup exam and provide any required documentation.
2. Students should assume that if approved, they will be scheduled for the Makeup exam that is listed in this course syllabus. If the student also has a conflict with the date and time of the Makeup exam, the student should *include that information in their initial makeup exam request.*

Deadlines for Makeup exam requests Requests submitted:

- by 11:59pm on the day *before* the Regular exam will be reviewed *before* the Regular exam begins.
- from 12:00am until 6:15pm on the day of the Regular exam may not be reviewed before the Regular exam begins.

Decisions made about your exam request

After the request has been reviewed, both the student and instructor will receive an email with one of the following:

- APPROVED – The email will include the date, time, and location of the Makeup exam. If the assigned Makeup time conflicts with another academic obligation, the department will assign an alternate time. Again, the student should *include any information about conflicts with the Makeup exam in the initial makeup exam request.*
- PENDING ADDITIONAL DOCUMENTATION – Additional documentation is required before a decision can be made. The student should assume the current request is denied and should *resubmit their request with the requested documentation.*
- DENIED – The student may still take the Makeup exam, but a penalty will be applied.

Students who have questions about the policy, wish to appeal a decision, or are unable to submit a request before 6:15pm on the date of the Regular exam should contact the course coordinator found at the end of this syllabus.

The Math Department offers a makeup exam for students who are unable to take the Regular midterm exams. The Makeup exam will typically be scheduled in the evening on the following day of the Regular exam, except Thursday exams. **Makeup exams are offered only once and will not be rescheduled.**

Instructions on Makeup Exam night. The student is responsible for knowing the room and time of the makeup examination. The instructions for the Regular exam will also be followed for the Makeup exam.

The makeup examination will be given from **6:15 – 7:30 p.m.** on the following dates:

		Deadline to sign-up to receive approval <i>before</i> exam	Deadline to sign-up on-line on the Math website	Deadline to request a make-up from the coordinator <i>after</i> the exam due to an emergency
Makeup Midterm 1	Wednesday, 9/30 6:15 – 7:30 p.m.	Tuesday, 9/29 11:59 p.m.	Wednesday, 9/30 6:15 p.m.	Thursday, 10/1 11:59 p.m.
Makeup Midterm 2	Monday, 11/9 6:15 – 7:30 p.m.	Wednesday, 11/4 11:59 p.m.	Thursday, 11/5 6:15 p.m.	Friday, 11/6 11:59 p.m.

CONFLICT FINAL EXAM POLICY There are two types of conflict final examinations: direct and overload. Direct conflicts are two exams scheduled at the same time. Overload exams are defined as three or more exams scheduled in consecutive time periods or within one calendar day. Students may elect to take the three or more exams on the same day if they wish or request a conflict final exam. **A student must take action to request a conflict exam through LionPATH between Monday, September 28 and Sunday, October 25. Final Conflict examinations cannot be scheduled through the Mathematics Department.** Students who miss or cannot take the final exam due to a valid and documented reason, such as illness, may be allowed to take a makeup final exam at the beginning of the next semester if they qualify for a deferred grade. Personal business, such as travel, employment, weddings, graduations; attendance at public events, such as concerts and sporting events; or forgetting the date, time, or room of the exam **is not a valid excuse**. If the student does not have a valid reason, a 30-point penalty will be imposed. Students in such a situation must contact their instructor within 24 hours of the scheduled final exam and all such makeup exams must be arranged through the course coordinator. Students who have taken the original final exam are not permitted to take a makeup final exam.

GRADING POLICY

COURSE GRADES: Grades will be assigned out of 100 percent, distributed as follows:

Midterm Exam 1	22.5%
Midterm Exam 2	22.5%
Final Exam	25%
Quizzes and Section Work	10%
MyLab Math Homework	10%
Written Homework	10%
Total	100%

Final course grades will be assigned as follows:

Grade	Percent Score
A	93% - 100%
A-	90%-92%
B+	87%-89%
B	83%-86%
B-	80%-82%
C+	77%-79%
C	70%-76%
D	60%-69%
F	0%-59%

The unavoidable consequence is that some students will be “just a point” away from the next higher or lower grade. For reasons of fairness, the policy in this course is to **NOT** adjust individual grades in such circumstances.

NOTE: Each student's letter grade will be based **EXCLUSIVELY** on the examinations, homework, quizzes, and section work. **There is no "extra-credit" work.**

DEFERRED GRADES: Students who are **currently passing a course** but are unable to complete the course because of illness or an emergency may be granted a deferred grade which will allow the student to complete the course within ten weeks after the final grade reporting deadline for the semester. Note that deferred grades are limited to those students who can verify and document a valid reason for not being able to take the final examination. For more information see [Deferred Grades](#).

GENERAL UNIVERSITY POLICIES

LATE-DROP: Students may add/drop a course without academic penalty within the first six calendar days of the semester. A student may late drop a course within the first twelve weeks of the semester. After the first six days and before deciding to late drop this course, each student should consult with their academic advisor. The late drop deadline is **Friday, November 13th**.

ACADEMIC INTEGRITY: According to Penn State policy G-9: Academic Integrity, an academic integrity violation is “an intentional, unintentional, or attempted violation of course or assessment policies to gain an academic advantage or to advantage or disadvantage another student academically.” Unless your instructor tells you otherwise, you must complete all course work entirely on your own, using only sources that have been permitted by your instructor, and you may not assist other students with papers, quizzes, exams, or other assessments. If your instructor allows you to use ideas, images, or word phrases created by another person (e.g., from Course Hero or Chegg) or by generative technology, such as ChatGPT, you must identify their source. You may not submit false or fabricated information, use the same academic work for credit in multiple courses, or share instructional content. Students with questions about academic integrity should ask their instructor **before submitting work**. **Students facing allegations of academic misconduct may not drop/withdraw from the affected course** unless they are cleared of wrongdoing (see G-9: Academic Integrity). Attempted drops will be prevented or reversed, and students will be expected to complete course work and meet course deadlines. Students who are found responsible for academic integrity violations face academic outcomes, which can be severe, and put themselves at jeopardy for other outcomes which may include ineligibility for Dean’s List, pass/fail elections, and grade forgiveness. Students may also face consequences from their home/major program and/or The Schreyer Honors College.

In order to ensure that all students have a fair and equal opportunity to succeed in this course, the Mathematics Department is committed to enforcing the University’s academic integrity policy. Below is a description of academic misconduct and the department’s responsibilities when misconduct is suspected.

Academic Misconduct

In this course, academic misconduct includes, but is not limited to:

- Copying the work of another student on an exam, quiz, or assignment;
- Passing off the work of another individual as your own;
- Using non-approved devices or aids, such as generative AI, on exams, quizzes, or assignments;
- Having unauthorized possession of exams or quizzes;
- Engaging in deception in order to extend or reschedule an exam, quiz, or assignment;
- Facilitating acts of academic misconduct by others.

When Academic Misconduct is Suspected

If a student is suspected of academic misconduct, the instructor’s duties are to:

- Confidentially inform the student of the allegation;
- Enter the charge and recommended sanctions on an Eberly College of Science Academic Integrity form;
- Proceed with the Academic Integrity charge process. The student can choose to accept or contest the allegation at this point.

Note that a student’s refusal to respond to the charges within a reasonable period of time is construed as acceptance of the allegation and proposed sanctions. Once the Academic Integrity form has been accepted or contested by the student, it is sent to the College’s Academic Integrity Committee for adjudication. A student cannot drop or withdraw from the course during the adjudication process.

Sanctions

If a student accepts an academic misconduct allegation, or if they are found guilty during adjudication, probable sanctions include:

- A warning and
- Reduction of the assignment grade to zero or
- Reduction of the quiz or exam grade to zero.

Additional sanctions might include:

- Reduction in the final course grade;
- An F in the course.

In addition, the student will be unable to drop or withdraw from the course.

Please see the [Eberly College of Science Academic Integrity homepage](#) for additional information and procedures.

STUDENTS WITH DISABILITIES: Penn State welcomes students with disabilities into the University's educational programs. If you have a disability-related need for reasonable academic adjustments in this course, contact Student Disability Resources at 814-863-1807 (V/TTY). For further information, please visit Student Disability Resources web site: <http://equity.psu.edu/student-disability-resources/>.

In order to receive consideration for accommodations, you must contact SDR and provide documentation (see the documentation guidelines at <http://equity.psu.edu/student-disability-resources/>). If the documentation supports your request for reasonable accommodations, SDR will provide you with an accommodation letter identifying appropriate academic adjustments. Please share this letter with your instructors and discuss the accommodations with them as early in your courses as possible. You must follow this process for every semester that you request accommodations.

CODE OF MUTUAL RESPECT AND COOPERATION: The Eberly College of Science Code of Mutual Respect and Cooperation pertains to all members of the college community; faculty, staff, and students. The [Code of Mutual Respect and Cooperation](#) was developed to embody the values that we hope our faculty, staff, and students possess, consistent with the aspirational goals expressed in the Penn State Principles. The University is strongly committed to freedom of expression, and consequently, the Code does not constitute University or College policy, and is not intended to interfere in any way with an individual's academic or personal freedoms. We hope, however, that individuals will voluntarily endorse the 12 principles set forth in the Code, thereby helping us make the Eberly College of Science a place where every individual feels respected and valued, as well as challenged and rewarded.

EDUCATIONAL EQUITY: The Office of the Vice Provost for Educational Equity serves as a catalyst and advocate for Penn State's diversity and inclusion initiatives. Educational Equity's vision is a Penn State community that is an inclusive and welcoming environment for all. If you wish to learn more or if you wish to [report bias](#), please visit the Educational Equity [website](#).

COUNSELING AND PSYCHOLOGICAL SERVICES: Many students at Penn State face personal challenges or have psychological needs that may interfere with their academic progress, social development, or emotional wellbeing. The university offers a variety of confidential services to help you through difficult times, including individual and group counseling, crisis intervention, consultations, online chats, and mental health screenings. These services are provided by staff who welcome all students and embrace a philosophy respectful of clients' cultural and religious backgrounds, and sensitive to differences in race, ability, gender identity and sexual orientation.

[Counseling and Psychological Services at University Park \(CAPS\)](#)

Penn State Crisis Line (24 hours/7 days/week): 877-229-6400

Crisis Text Line (24 hours/7 days/week): Text LIONS to 741741

QUESTIONS, PROBLEMS, OR COMMENTS: If you have questions or concerns about the course, please consult your instructor first. If further guidance is needed, you may contact the course coordinator whose contact information is given below.

Darla Forsha Simpson

Associate Teaching Professor

403 McAllister Building

University Park, PA 16802

Email: math220coord@psu.edu or dff3@psu.edu

Include your **Name, Student ID, and Section Number** in any correspondence and always use your @psu.edu email address.